

- Q.27 Explain the open-circuit and short-circuit tests on transformer.
- Q.28 Explain different types of losses in a transformer and their minimization.
- Q.29 Write down the condition for parallel operation of transformers.
- Q.30 Describe the operation of an autotransformer.
- Q.31 Discuss different types of DC motors and their applications.
- Q.32 Differentiate between core and shell type transformers.
- Q.33 Write down the differences between power transformer. and distribution Transformer.
- Q.34 Explain the application of three-phase transformer.
- Q.35 Explain the various types of DC generator using necessary equations and circuit diagram.

SECTION-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x10=20)

- Q.36 Explain the performance characteristics of DC series and DC shunt motor.
- Q.37 Derive condition for maximum efficiency of a transformer.
- Q.38 Write down the constructional details of various parts of DC machines with neat diagram?

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**4th Sem / Elect, GE, Power Station Engg., Elect.
& Eltx. Engg., Fire Tech. & Safety
Subject:- Electrical Machines - I**

Time : 3Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 The device which converts electrical energy into mechanical energy is called-
- a) Transformer b) Generator
- c) Motor d) Converter
- Q.2 The energy conversion device that does not rotate is-
- a) Generator b) Transformer
- c) Motor d) Alternator
- Q.3 The armature winding is placed on-
- a) Stator b) Rotor
- c) Yoke d) Pole shoe
- Q.4 The type of winding suitable for high current, low voltage is-
- a) Lap winding b) Wave winding
- c) Ring winding d) Series winding

- Q.5 The material used for brushes is-
- Copper
 - Iron
 - Graphite
 - Aluminum
- Q.6 The compound wound DC motor has-
- Only series field
 - Only shunt field
 - Both series and shunt field
 - No field winding
- Q.7 The function of transformer oil is-
- Cooling and insulation
 - Lubrication
 - Flux creation
 - Voltage control
- Q.8 Open circuit test is used to find-
- Copper losses
 - Iron losses
 - Mechanical losses
 - Efficiency
- Q.9 The cooling method most commonly used for large transformer is-
- Air cooling
 - Oil natural
 - Water cooling
 - Air blast
- Q.10 The condition for parallel operation of transformer is-
- Same voltage ratio
 - Same polarity
 - Same impedance
 - All of these

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SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 What is the function of the commutator in a DC machine?
- Q.12 What is an electrical machine?
- Q.13 Define commutation.
- Q.14 In lap winding number of parallel paths=_____
- Q.15 What is a lap winding?
- Q.16 Define efficiency of a transformer.
- Q.17 What is an ideal transformer?
- Q.18 Why core of transformer is laminated?
- Q.19 Name two types of losses in transformers.
- Q.20 What is an autotransformer?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Discuss losses in electrical machines and their effects.
- Q.22 Describe and draw the equivalent circuit of single phase transformer using necessary equations and circuit diagram.
- Q.23 Derive the emf equation of a DC generator.
- Q.24 Explain the characteristics of a DC shunt motor.
- Q.25 Discuss armature reaction and its effects.
- Q.26 Explain the speed control methods of DC motors with diagrams.

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